FEDERICO DI MASE

Mechanical Engineering Graduate

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PROFESSIONAL PROFILE

Mechanical Engineering graduate with hands-on experience in motorsport, vehicle design, structural analysis, and experimental testing. Formula Student sub-team leader and competitive motorcycle racer with trackside experience supporting a professional race team during a WorldSBK Championship weekend. Experienced in CAD, FEA, engineering data analysis, mechanical integration, and practical vehicle preparation.

MOTORSPORT EXPERIENCE

Trackside Motorsport Support & Data Analysis **09/2026**

ARCO Yamaha MotoR University Team *WorldSBK Championship Weekend*

- Supported the team throughout a full WorldSBK Championship weekend, assisting with Yamaha R3 and R7 motorcycle preparation, maintenance, and trackside mechanical duties.
- Gained practical race-engineering experience by supporting engineers with vehicle data analysis and observing the process of evaluating motorcycle performance and setup changes during live competition.
- Analysed motorcycle telemetry and lap-performance data using **RaceStudio3**, identifying trends in rider and vehicle behaviour for discussion with the engineering team.
- Analysed **lambda data using MeCal** to support evaluation of engine and fuelling behaviour.
- Assisted with suspension setup activities and gained practical exposure to **ZeroChassis** chassis and suspension setup software.

Sub-Team Leader – Powertrain-Battery **01/2025 – 09/2025**

Manchester Stinger Motorsport *Manchester, England*

- Led the Powertrain-Battery sub-team responsible for mechanical integration and packaging of the battery system within a fully electric Formula Student vehicle.
- Delivered the team's mechanical design and manufacturing programme **two weeks ahead of schedule**, while meeting racing and structural requirements.
- Designed CAD layouts, mounting solutions, and mechanical interfaces within tight chassis constraints and required clearances.
- Developed manufacturable mounting solutions designed to withstand racing loads of up to **40g**.
- Promoted from Battery Sub-Team Member to Sub-Team Leader based on contribution to the project.

ENGINEERING EXPERIENCE

Junior Production Engineer **06/2026 – Present**

Adarte SRL *Apricena, FG, Italy*

- Conduct time-and-motion studies on shop-floor operations, establishing accurate cycle times and identifying opportunities for process improvement.
- Reverse-engineer physical components in **SolidWorks**, developing improved designs and assessing opportunities for in-house manufacture.

SELECTED ENGINEERING PROJECT

Design & Structural Validation of a Hybrid Vehicle Frame **09/2025 – 05/2026**

University of Manchester – Individual Engineering Project

- Designed and validated a hybrid vehicle frame using **SolidWorks** and **ANSYS Mechanical**, combining CAD development, experimental testing, and numerical analysis.
- Established experimentally validated material properties and used them in structural simulations to assess frame performance against applicable ISO safety requirements.
- Conducted compressive and 4-point bending tests on bamboo, processing multi-axis material data using Excel.
- Performed static structural, modal, and eigenvalue buckling analyses with mesh convergence studies to evaluate structural performance and design feasibility.

EDUCATION

BEng (Hons) Mechanical Engineering – 2:1 **2023 – 2026**

University of Manchester *Manchester, England*

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

Motorcycle Racing: Yamaha R3 Cup Italy Rider (2022–2023) • CNV Motorcycle Regional Champion (2020)
Leadership: Formula Student Sub-Team Leader • Student Body President (2022–2023) • University Ambassador (2025–2026)
Award: Hack-a-bot Sony Competition – 2nd Place (2025)
Internship: Adarte SRL – Summer 2024 & 2025
Languages: Italian (Native) • English (Professional Working Proficiency)
Licences: Driving Licence B, A2